

04/06/26

DAY 09 "Kamineni Dhana Sri Keerthi"

working on

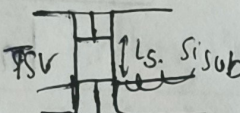
Q1 Defected TSV Electrical Modelling

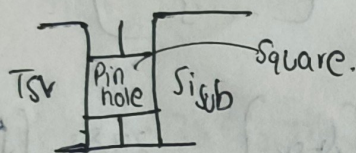
1.1 Pinhole defect is Modelled in terms of RL

1.2 Pinhole defect is Modelled in terms of C_{leak} & G_{leak} .

1.1 Pinhole defect [R, L terms]

1.1.1 what is the shape of pin hole defect?

A Square pin hole. 



1.1.2 what is the material is in the pinhole?

A Not air, a conductive channel and represent its resistivity as $\rho_T = \rho_0 [1 + \alpha \cdot (T - T_0)]$.

1.1.3

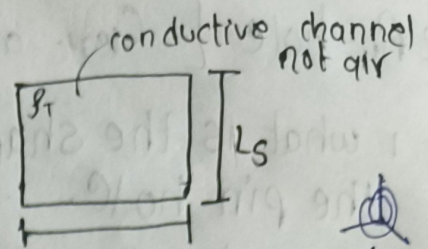
$$R_{shut} = \frac{\rho_T t_{ins}}{L_s^2} \quad \begin{matrix} \text{current path length} \\ \text{depth or thickness} \\ \text{I have to cross} \end{matrix}$$

L_s^2 [cross sectional area]

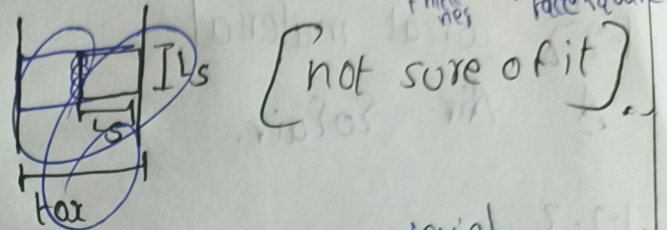
$\rho_T \rightarrow$ resistivity of conductive channel in the pinhole.

$t_{ins} \rightarrow$ thickness of insulator or oxide layer or SiO_2

$L_s \rightarrow$ side of square.



if $L_s \neq t_{ox}$ sometimes



1.1.4

$$L_{shc} = \frac{\mu \mu_{TSV} t_{ins}}{2\pi} \left[\cosh^{-1} \left(\frac{t_{ins}}{\sqrt{L_s^2 + t_{ins}^2}} \right) \right]$$

main field contribution

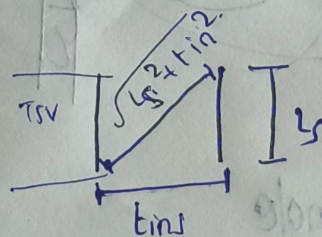
$$- \sqrt{2 \cdot t_{ins}^2 + L_s^2} + \frac{t_{ins}}{a} + \sqrt{L_s^2 + t_{ins}^2}$$

square shape.

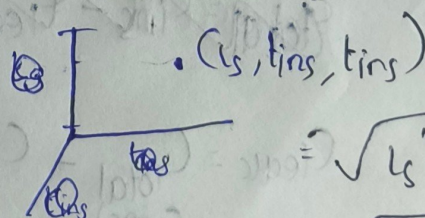
$L_{shc} =$ main magnetic field contribution - finite geometry corrections.

$$\text{main field} = \frac{\mu \mu_{TSV} t_{ins}}{2\pi} \left[\cosh^{-1} \left(\frac{t_{ins}}{\sqrt{L_s^2 + t_{ins}^2}} \right) \right]$$

mfield integration $\rightarrow$ Rosadyle logarithmic



$$= \sqrt{L_s^2 + t_{ins}^2}$$



$$= \sqrt{L_s^2 + t_{ins}^2 + t_{ins}^2}$$

$$= \sqrt{2 t_{ins}^2 + L_s^2}$$

$I_{main} - I_{main-sq\ element}$ magnetic coupling between

inductor

current paths

1.2 Pin hole [C_{leak} and G_{leak}]

1.2.1 what is the shape of the pin hole.

A Cylindrical

1.2.2 what material.

A Air $\epsilon_0 \epsilon_{air}$.

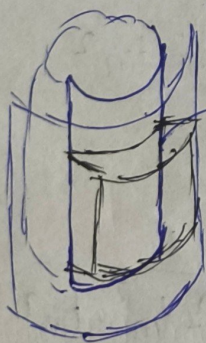
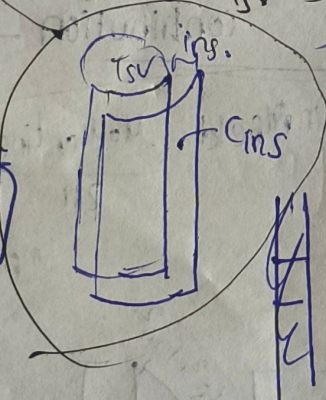
1.2.3

$$C_{leak} = \left(\frac{2\pi \epsilon_0 \epsilon_{ox} l_{TSV}}{\ln \left(\frac{r_{TSV} + t_{ox}}{r_{TSV}} \right)} \right) -$$

$$\left(\frac{2\pi \epsilon_0 \epsilon_{air} j r_{TSV}}{\ln \left(\frac{r_{TSV} + t_{ox}}{r_{TSV}} \right)} \right)$$

general:

$$C = \frac{2\pi \epsilon l}{\ln \left(\frac{b}{a} \right)}$$



pin hole

$$C_{total} = C_{ins} - C_{leak}$$

$$C_{leak} = C_{total} - C_{pinhole \text{ or } air}$$

j = damaged area

side wall area affected

$$= \frac{\text{angular extent of pinhole}}{2\pi r_{TSV}} = \frac{wh}{2\pi r_{TSV}}$$

arc length & air type of

$$C_{leak} = C_{ins} - C_{air}$$

capacitance lost due to defect.

1.2.4 G_{leak} :

$$2\pi \epsilon_0 \epsilon_{ox} [l_{TSV} - j r_{TSV}]$$

$$\ln \left(\frac{r_{TSV} + t_{ox}}{r_{TSV}} \right)$$

$$= \frac{2\pi \epsilon_0 \epsilon_{ox} [l_{TSV}]}{\ln(x)} - \frac{2\pi \epsilon_0 \epsilon_{ox} [j r_{TSV}]}{\ln(x)}$$

$$j = G E$$

$$G = j\omega C$$

$$j = \frac{\omega h}{2\pi r_{TSV}}$$

$$f = \frac{\omega}{2\pi} \quad \omega = 2\pi f$$

$$G = j 2\pi f C$$

1 TSV | pinhole | silicon
 $\downarrow \uparrow$ so G_{leak}

$$Z_{leak} =$$

$$\frac{2\pi\epsilon_{SiO_2} l_{TSV}}{\ln\left(\frac{r_{TSV} + l_{ox}}{r_{TSV}}\right)} - \frac{4\pi^2 j f \epsilon_0 \epsilon_{air} r_{TSV}}{\ln\left(\frac{r_{TSV} + l_{ox}}{r_{TSV}}\right)}$$

or

$$\frac{2\pi\epsilon_{SiO_2} (l_{TSV} - j r_{TSV})}{\ln(x)} - \frac{2\pi\epsilon_{air} (l_{TSV} - j r_{TSV})}{\ln(x)}$$

$$\frac{2\pi\epsilon_{SiO_2} [l_{TSV}] - 2\pi\epsilon_{SiO_2} j r_{TSV}}{\ln(x)}$$

$$- \frac{2\pi\epsilon_{air} [l_{TSV}] + 2\pi\epsilon_{air} j r_{TSV}}{\ln(x)}$$

G_{leak} = oxide conductance -
 $\swarrow$ air capacitive admittance

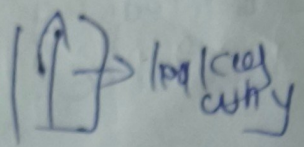
$$\frac{2\pi\epsilon_{SiO_2} [l_{TSV}]}{\ln(x)} - \frac{4\pi^2 j f \epsilon_0 \epsilon_{air} r_{TSV}}{\ln(x)}$$

$$Y_{air} = j\omega C$$

$$= j 2\pi f \left[\frac{2\pi \epsilon_0 \epsilon_{air} r_{TSV}}{\ln(x)} \right]$$

conducⁿ of air ≈ 0 but since
 it is leaking they took
 admittance:

1 why use distributed model

2 TSV signa  logical why

3 diff. types of insulators?

4 write in ppt

case i) Cu leakage

case ii) air leakage

RLC model.

5 Other kind of leakages.

Basic comparison parameters.

→ not week Saturday.

delay
crosstalk,
noise.

Shape → cylinder

lumped model → low freq.

distributed model → High freq.

In matlab make

all equations

defect free,

defected.

Today Novel RL

same sum $\left\{ \begin{array}{l} \text{parameters} \\ \downarrow \\ \text{same for all} \end{array} \right\}$

data table.

→ madam reference code.

12:30 pm